

Lab 7 - Ohm's Law

Objectives

1. **Design a step-by-step experiment procedure** to study Ohm's law

Introduction

Ohm's law states that:

$$V = IR \quad (1)$$

Where V, I, and R represent the voltage in volts (V), current in amperes (A), and resistance in ohms (Ω), respectively, for a resistor or segment of DC circuit. R is the equivalent resistance if the segment consists of multiple resistors.

Equipment

- Resistors
- 2x TENMA 72-7720 Multimeter
- Amprobe CR50A Multimeter (Optional)
- DC Power supply
- Breadboard
- Connecting wires

Experiment Requirements

Build a circuit according to the diagram in figure 1 using resistors with nominal values, $R1 = 100\Omega$, $R2 = 220\Omega$, $R3 = 1k\Omega$, $R4 = 2.2k\Omega$, and $R5 = 1k\Omega$. A power supply should be connected to points A and B. Vary the voltage setting on power supply from 10 to 20 V with at most 1 V increment. Choose 2 different components from List 1 and 2 different segments from List 2. Each group member must pick one of these components/segments then measures the relevant variables to produce a data table and/or plots to show if the Ohm's law is satisfied (or not) within the measured range. If there are less than 4 members in your group, you can pick more than one component/segment. You can also pick more components/segments if you want to do more analysis.

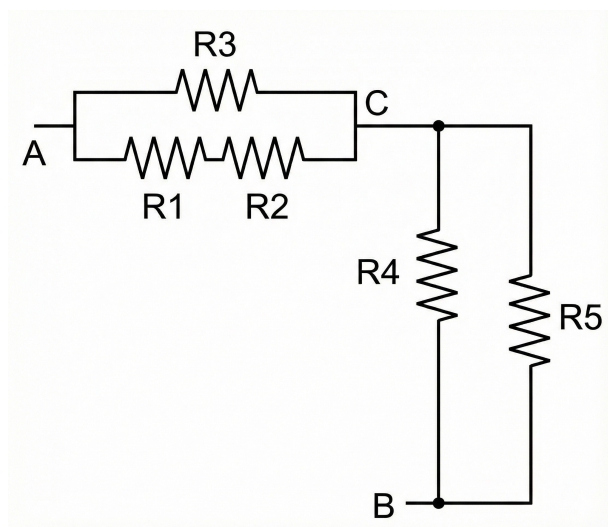


Figure 1: Ohm's Law Circuit Diagram

List 1

Components:
R1, R2, R3, R4, R5

List 2

Segments:
AB, AC, CB

Precaution

- **Pick a dedicated multimeter as an ammeter and fix its scale to 200 mA.** This will avoid a multimeter from switching between ammeter and voltmeter, which can easily blow the fuse.
- **Use the other multimeter when measuring resistance and voltage.**
- **Before building the big circuit,** first measure and note the actual resistance of each component pulled from the cabinet in a preliminary data table, remembering to keep track of units and scales used. Remember to record the instrument uncertainty for each.

Lab Report Requirements

1. **Special requirement of the week:** State your procedure clearly and completely.
 - The procedure should be detailed enough for someone else to repeat the experiment and get the same result.
 - You may use a flow chart, a step-by-step list, or any other format you like as long as it is clear and complete.
 - Use the, "If my lab partner were woken up at 3 am and asked to do the experiment for the first time, could they do it easily?" as a test for your procedure.
2. Describe the data analysis including uncertainty analysis
3. Include enough details in your data table(s) and plots to support your conclusion
4. Cite your data and plots to support your conclusion

Hints for Proving Ohm's Law

These are two common approaches, but you are not limited to them. For whichever approach you pick, make sure you understand the uncertainties involved and how they are propagated to the final result. You

must include the uncertainty analysis in your lab report to support your conclusion.

1. You can use your measured R and I to calculate V , then compare it with measured V
2. You can fit V vs. I plot with linear function, then compare the slope (should be R) with the measured R .

Pre-Lab Questions

1. Detail your experiment procedure:
 - (a) Choose 2 different components from List 1 and 2 different segments from List 2 as required in lab instruction
 - (b) Describe your data collection, analysis, and plotting procedure
 - (c) Describe and show how you will analyze the uncertainties and propagate them to the final result
 - (d) How can you tell if the Ohm's law is satisfied or not based on your chosen approach?

Appendix A - Tenma 72-7720 Multimeter Instrument Uncertainties

Refer to the multimeter manual for any additional information.

Note: lsdp = least digit place

DC Voltage

Scale	Instrument Uncertainty
All Scales	$\pm(0.5\% \text{ rdg} + 1 \text{ lsdp})$

DC current

Scale	Instrument Uncertainty
2 mA	$\pm(0.8\% \text{ rdg} + 1 \text{ lsdp})$
200 mA	$\pm(1.5\% \text{ rdg} + 1 \text{ lsdp})$
20 A	$\pm(2.0\% \text{ rdg} + 5 \text{ lsdp})$

Resistance

Scale	Instrument Uncertainty
200 Ω	$\pm(0.8\% \text{ rdg} + 3 \text{ lsdp})$
2k Ω – 2M Ω	$\pm(0.8\% \text{ rdg} + 1 \text{ lsdp})$

Appendix B - Amprobe CR50A Multimeter Instrument Uncertainties

Refer to the multimeter manual for any additional information.

Note: lsdp = least digit place which is the same as dgt (digit) in the manual.

Resistance

Scale	Instrument Uncertainty
20 Ω	$\pm(1.2\% \text{ rdg})$
200 Ω	$\pm(0.5\% \text{ rdg} + 3 \text{ lsdp})$
2k Ω to 2M Ω	$\pm(0.5\% \text{ rdg} + 1 \text{ lsdp})$
2M Ω	$\pm(3.0\% \text{ rdg} + 1 \text{ lsdp})$

Capacitance

Scale	Instrument Uncertainty
200pF	$\pm(0.5\% \text{ rdg} + 1 \text{ lsdp} + 0.5\text{pF})$
2000pF to 2 μ F	$\pm(0.5\% \text{ rdg} + 1 \text{ lsdp})$
20 μ F	$\pm(0.5\% \text{ rdg} + 1 \text{ lsdp})$
200 μ F	$\pm(0.5\% \text{ rdg} + 1 \text{ lsdp})$
2mF	$\pm(1.0\% \text{ rdg} + 1 \text{ lsdp})$
20mF	$\pm(1.5\% \text{ rdg} + 1 \text{ lsdp})$